

## Site conditions have an impact on compensation payments for the loss of tree species in mixed forests

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Selective browsing of saplings is an unavoidable consequence of high density of wild game and can lead to stand homogenization. To ameliorate the effects of economic risks associated with decrease in species diversification – for example, decreased flexibility in relation to the timber market or increased susceptibility to calamities – compensation payments can be helpful. The risk components in forest economics are now well described. However, even though soil and climate conditions are among the most important factors determining financial gain or loss from forest management, differences due to variations among forest sites are rarely considered in economic analyses. In order to consider the impact of site variation on the financial effects of game browsing, this study focused on financial results of forest stand models considered representative for average site conditions for each of the 15 growth regions in Bavaria, Germany. Using the Silva 2 growth model, the average growth of spruce (*Picea abies* (L.) Karst.) and beech (*Fagus sylvatica* L.) was simulated to consider return–risk ratios of mixed and homogenized stands – thus to compare target and deviating states. The investigation was divided into two parts: first, as many forest decisions are dependent on financial outcomes, the analysis of the financial impact of varying site conditions began by calculating the annuities. Second, it was important to see what influence site conditions might have on compensation payments. The results showed that fluctuations in spruce growth rates among 15 growth regions of Bavaria were much higher than those for beech. Because of the loss of beech, when calculating necessary risk compensation levels, the results were largely dependent on the variations in beech growth and the financial risk level of spruce – both due to differences in site quality.

### Introduction

The impact of ungulates on vegetation is well documented (e.g. Gill, 1992; Horsley *et al.*, 2003; Husheer *et al.*, 2003). In addition to growth loss or reduction in timber quality, recruitment failure of admixed saplings can be considered a management risk for the forest owner. While methods are available for evaluating financial loss due to growth loss and reduction in timber quality (e.g. Pilz, 1905; Kroth *et al.*, 1985; Ward *et al.*, 2004), Clasen *et al.* (2011) pointed out the lack of methods for evaluating the financial impacts of homogenization of tree species for which the forest owner himself is not responsible. They also demonstrated how to derive the financial compensation that would be necessary to make acceptable the increased risk for losing admixed tree species. In their examples, potential recruitment failure of 30 per cent of the original beech (*Fagus sylvatica* L.) component of different mixtures with spruce (*Picea abies* (L.) Karst.) would require annual payments between 16 and 23 € ha<sup>-1</sup> to compensate for the higher risk of a single stand. Hereby, the risk is defined as deviation from the mean annual financial return, expressed as the standard deviation.

In such economic evaluations, complex models that consider fluctuating timber prices and calamities such as bark beetle

infestation or wind damage are used. It should be noted, however, that the impact of site conditions is often ignored, and therefore, any regional variations in site class are generally not considered. However, this assumption of constant site conditions has been criticized (Lopez *et al.*, 2010). This criticism is not without merit, since factors such as local humidity and nutrients in the soil also have a strong influence on the growth potential of tree species in regions across which climate is fairly uniform (Ellenberg and Leuschner, 2010). In addition to spatial differences, temporal aspects should also be considered. In this case, Skovsgaard and Vanclay (2013) are recommending adaptive modelling of site-specific parameters. Thus, long-term planning for a forest stand can no longer be carried out using an unchanging site factor, due to expected future environmental changes (Bredemeier *et al.*, 2001).

As noted, in spite of the importance of these site variations, Clasen *et al.* (2011) used site conditions from only one growth district as a basis for their calculations. Thus, possible deviations in economic yields due to varying site conditions were not considered. However, the assumption of constant site conditions is widely used in many economic studies. For example, Hanewinkel *et al.* (2010) described the changes of land expectation value (LEV) across biome shifts for spruce and beech for a forest area of 1.3 million

hectares in Baden-Württemberg, Germany, and assumed equal site conditions for the entire federal state. The same assumption was made by Dieter (2001) for one specific region in southern Germany when calculating LEV for spruce and beech. Here, the question that the author posed was whether or not beech achieved higher LEV than spruce when survival probabilities were considered. Although spruce showed lower survival probability, this species was still more profitable. However, Dieter (2001) simulated spruce growth with excellent conditions and limited the results only to regions with comparable soil and climate conditions. Indeed, the opposite conclusion can be reached when one assumes unfavourable conditions for spruce instead. Other authors also came to the same conclusion. Garcia-Gonzalo *et al.* (2007) described the financial results for a specific study area and pointed out that one of the weaknesses of their model is that the results cannot be extrapolated to other regions.

### State of the art: the impact of varying site conditions

Site variations are well documented. Since the end of the eighteenth century, yield tables based on the results of experimental plots have been used (Wiedemann, 1932; Assmann and Franz, 1963), which show the increment and growing stock of tree species on sites of various qualities (Kennel, 1971). A growing interest in yield-based processes resulted in the design, implementation and description of several forest yield investigations. In Central Europe, research about the main tree species – spruce and beech – predominates. For example, Kennel (1965) showed the increment and production relationships for beech and spruce for three sites. Von Lüpke and Spellmann (1997) as well as Pretzsch (2003) demonstrated that the growth relationship between these two tree species varies from site class to site class. With increased experience and the additional ability to indicate and predict forest growth using simulation models provided by computer-aided systems, this research obtained a new dimension (Pretzsch, 2000). Recent studies have also described an increase in the average annual increment correlated with higher nitrogen and carbon dioxide emissions (Pretzsch, 1999, 2009). In this case as well, the result is dependent upon the site class.

Brandl (1988, 1989) described two approaches for analysing the profitability of tree species under different site qualities. The first was based on statistical data: comparing net income of forest enterprises with similar tree species composition was used to differentiate between regional site qualities. But Brandl (1988) himself criticized this approach, due to the fact that usually, seemingly similar enterprises have, in fact, marked differences due to forest size, age-class distribution or price–cost structures, therefore drawing conclusions about the influence of site quality variations is obviously problematic. The second approach was based on model calculations: Brandl (1988, 1989) showed step-wise changes in net income when changing the growth performance of tree species – thus effectively simulating differences in site quality. However, specific relationships to actual regions – e.g. based on inventory data – were still missing. Based on a similar approach, Kroth (1983) calculated revenues and costs, which were dependent upon production capacity for pine (*Pinus sylvestris* L.). For this, he used yield table data and looked at the financial consequences of different yield classes. Similarly, Ripken and Spellmann (1980) calculated net income for the main tree species in Lower Saxony, Germany. They also used data from

yield tables and estimated the second yield class as an average of all forest sites. Thus, only the growth differentials between the tree species were recognisable. Further studies investigated whether a varying level of quality of site had an effect on the optimal harvest strategy (Marutani, 2010). Finally, Halbritter and Deegen (2011) found that established valuation methods like the Faustmann approach can lead to misinterpretation if soil degradation is not considered.

Obviously, as the literature illustrates, variation in yield class can lead to a broad range of economic results. This range exists – to a greater or lesser extent – for every tree species. Based on this background, the aim of this study is not to show small-scale site variations, as resulting financial variations seem to be obvious; rather, it is intended to show financial differences between average site conditions for different growth regions; as, although studies often consider large regions as a basis for economic valuations, thus far no study has integrated growth differences comprehensively. Going further, it was looked at how these site variations can influence the necessary risk compensation payments as calculated by Clasen *et al.* (2011). Research about ungulate effects and their economic impacts is particularly relevant in the present, as in Central Europe, for example on-going discussions between forest owners and those possessing hunting rights on forest land must be firmly grounded by scientific arguments. Hereby, the scientific methods used must include growth site variations.

To conclude, following questions were addressed:

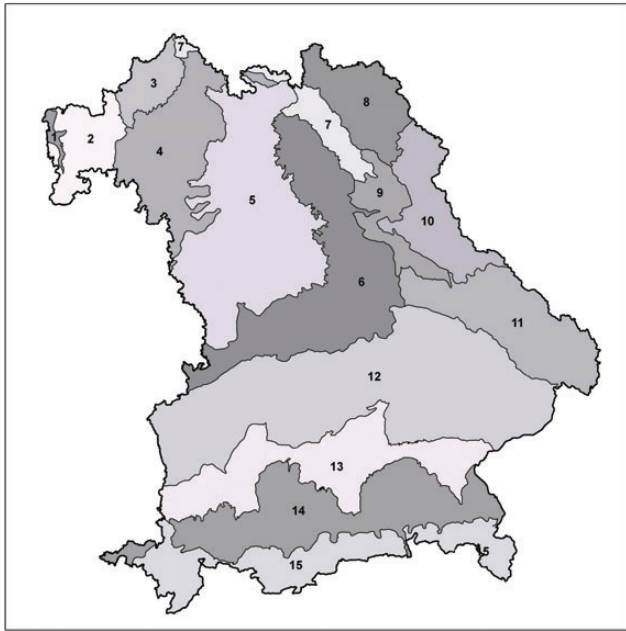
- To what extent will the financial results of timber management change from growth region to growth region, each with its own average soil and climate parameters (i.e. site conditions), if all other parameters and management activities remain the same?
- How will the payments required to compensate for ungulate browsing fluctuate if these financial calculations are based on different site conditions?

This paper describes a simulation study based on actual inventory data. To answer the research questions, two steps were carried out. Firstly, the total volume production of both spruce and beech, one stand considered representative of each growth region of Bavaria (*ceteris paribus* analysis), was simulated to show variances in revenue due to site conditions. Secondly, it was checked to what extent site variations influenced the level of risk payments. Within this scope, the results were validated and discussed.

## Methodology and data

### Area under investigation

Fifteen growth regions, subdivided into 70 growth districts, have been classified for Bavaria (see Figure 1 and Table A1). They range in size between 18 000 and 1.4 million hectares, and total >7 million hectares (Kölling, 2005). The amount of land covered by forest in each growth region varies from 19 to 66 per cent. Elevations range from 150 to 2250 m above sea level – thus, the topography ranges from colline–submontane to alpine. The average yearly temperature of the forest areas varies between 5.2 and 8.6°C, with precipitation between 713 and 1720 mm per year, depending upon altitude (Kölling, 2005). The Bavarian forest area is composed of 45 per cent spruce, 19 per cent pine, 12 per cent beech and 6 per cent oak (*Quercus robur* L. and *Quercus petraea*



**Figure 1** Growth region distribution of Bavaria, Germany (reprinted with permission from C. Kölling, LWF).

(*Mattuschka*) Liebl.) with the remainder comprised mainly of mixed hardwoods.

### Growth simulation

The results of forest economic evaluations depend largely on biophysical data, such as thinning amounts and standing volume, for each period under investigation (for example, 5- or 10-year thinning intervals). For the purpose of economic evaluation, net revenues from thinnings and standing volume for each of these periods were calculated. This approach allowed us to compare the economic consequences of actions such as different silvicultural treatments or events such as biotic and abiotic damages.

For our survey, the single-tree-based growth simulation model *SILVA* 2 (Pretzsch et al., 2002) was used, which best reflects the growth situation in Bavaria. *SILVA* is capable of generating and projecting the growth of forest stands through to the phase of regeneration after harvest. The specific growth process for each growth district is determined by site-sensitive age–height curves, which are dependent on nine specific site parameters. In addition to these site factors, necessary input variables consist of mean diameter and basal area of the stand. In the course of simulation, the distance-dependent model considers each individual tree of the forest stand in a three-dimensional analysis. Functions for competition, thinning, increment and mortality all influence the growth of each tree. The growth prognosis is carried out in 5-year cycles (Pretzsch et al., 2002).

In this study, the main species spruce and beech for all growth regions were used, since both are present in nearly the entire federal state of Bavaria. Pretzsch (2002) tested the application of *SILVA* 2 using growth districts of Bavaria as representatives for larger regions. This was done using yield-related inventory data of representative 30-year-old forest stands located at the most

frequent site types for each growth region. Younger stands were not used, as the model shows good parameterization for growth only from older age-classes upwards (Hanewinkel, 2001). This procedure was followed and the simulation began at year 25 for spruce and 45 for beech.

The individual growth patterns in the central Alps region in southern Bavaria, however, cannot be simulated using *SILVA* 2, because occurrences such as rockfall and snow gliding on steep hillsides – which can influence the growth of trees – are difficult to integrate into the growth simulator. Thus, only the less hilly growth district of *Oberbayerische Flysch-Voralpen* as a representative of all less-affected growing areas of the Alpine region was considered. For all other growth regions, with more than one district, one representative district was randomly chosen as well.

The average growth for both tree species in the respective growth regions was achieved as a rough estimation: the start parameter for the simulation for each region was adapted individually, as it can be assumed that differentiation of the stand growth takes place early on. Based on data from the National Forest Inventory (NFI 2 (BMELV, 2004)), obvious differentiation of stand parameters (basal area, diameter and timber volume) was found in younger age-classes among the different growth regions. As these data were not sufficient for transferring into *SILVA* due to statistical uncertainty for the small regions, data from yield tables – Assmann and Franz (1963) for spruce and Wiedemann (1932) for beech – were used uniformly. Nevertheless, to produce data variation (diameter and basal area per hectare), the differentiation of the NFI data was used (BMELV, 2004) to adapt the yield table data. The data used thus constitute an approximation of an average forest stand for each of the growth regions for year 25 (spruce) and year 45 (beech). From this point on, specific age–height curves contained within *SILVA* 2 determined the individual growth of trees.

A 100-year rotation period for spruce and a 120-year rotation period for beech were assumed. The harvest amount was limited to  $100 \text{ m}^3 \text{ ha}^{-1}$  (gross volume under bark) and decade for both species. In total, 90 simulations were carried out – in 15 growth regions for both spruce and beech in a triplicate repetition for each species and region. Average volume values for each species, each period and each growth region were calculated from the replicates; thus the implemented growth stochasticity of *SILVA* 2 was taken into consideration. In the final step, these values were fed into the grading program *BDAT* (Kublin and Scharnagel, 1988) in order to divide the timber into common product grades for economic valuation.

### Modelling economic performance

The impact of varying forest site conditions on economic yield can be shown with financial parameters. Hereby, net revenues for each period – stumpage and thinning values minus harvesting costs – form the basis for calculation. To evaluate economic returns for the timber volume harvested, timber price data from 2007, which were provided by the State Forest Enterprise, were adopted (BaySF, 2007). Doing this, a mixed price for each diameter class from the various timber qualities was used. Taking into account costs for logging, net revenues for thinnings were calculated, and a stumpage value for each period simulated was derived. After 20 years for spruce and 30 years for beech, one-time tending costs of 500 € were considered for each stand. Furthermore, an existing understory at the beginning of

the simulation and a clear cut at the end of the respective rotation time were assumed.

Due to the time value of money, it is not possible to compare payments from different forest stands that occur at different times. The net present value (NPV) converts, by means of discounting, all future financial flows into present values, and thus provides a current value figure that considers potential revenues during the entire investment period (Kruschwitz, 2005). By following this procedure, investment options can be directly compared with one another (equation 1). An interest rate of 2 per cent was used, since this rate is appropriate considering the modest growth and long rotation periods typical of German forests (Bright and Price, 2000; Benítez et al., 2007). A sensitivity study with varying interest rate was not carried out, since the interest rate was not expected to have a considerable impact on the relative differences between the growth regions.

$$NPV = \sum_{t=0}^T p_t(1+i)^{-t} \quad (1)$$

with  $T$  being the planning horizon,  $t$  the considered point in time,  $p_t$  the net revenues at a given point in time and  $i$  being the decimal interest rate.

When using this method, the following must be considered: firstly, a positive final NPV value means a worthwhile investment, while a negative value indicates that the internal rate of return is lower than the interest rate. Secondly, a rational investor would invest in the opportunity with the highest NPV.

Since different rotation periods for spruce and beech were used, the NPV calculation had to extend using an annuity, which can be understood as making an annual profit contribution as long as it is positive (Möhring et al., 2006). For this, discontinuous payment flows are converted into one consistent annual payment amount by multiplying the NPV by the annuity factor:

$$a = NPV \cdot \frac{i(1+i)^T}{(1+i)^T - 1} \quad (2)$$

Calculating the annuity for the different forest stands – as a result of simulated timber volume, which is then valued with costs and prices – enables us to identify the differences between growth regions. Showing the financial differences per year and hectare between two or more stands is thus easily understood (Möhring and Rüping, 2008).

### Calculating risk payments due to ungulate browsing

The goal here, however, was not only to show the consequences of site variation on financial outcomes but also to see what influence variation among different growth regions could have on necessary risk compensation, for example, for game browsing. Economic methods such as those used to evaluate the impacts of a homogenization of tree species (Clasen et al., 2011) are also based on calculating NPV and annuities but are more complex, since risk components must be integrated. Profits from a forest can suddenly decrease due to unexpected occurrences: both local and global timber markets determine the timber price – whereby fluctuations of prices can lead to increased risk (Brazee and Mendelsohn, 1988; Knoke et al., 2001). Timber prices are particularly influenced by instances of widespread calamities – resulting in a sudden

increase in timber, with no corresponding increase in demand. In addition, calamity timber often has less value, as it is both devalued due to breakage and harvesting costs are higher (Bright and Price, 2000). The financial risk of such calamities is addressed in several studies (e.g. Hanewinkel et al., 2008 or Beinhofer 2008; Knoke et al., 2008).

To simulate uncertainties from fluctuating timber prices, data from the Bavarian State Forest Administration (BayStMELF, 2004) and the State Forest Enterprise (BaySF, 2007) recorded between 1975 and 2007 were used. Using a stochastic process, it was extrapolated from the 32 years of timber price data into the future. A price distribution for saw log-size classes, which have the highest timber volume, was used: for spruce resulting prices ranged between 25 and 29 cm and for beech between 35 and 39 cm. With these distributions, it was possible to calculate historical mean prices and their standard deviation for every site class for each of the respective tree species. In order to repeat the timber price simulation under different price scenarios, the net revenues were multiplied by a quotient, as described by Knoke et al. (2005). This quotient is simply the ratio of the simulated stochastic timber price and the mean historical timber price.

In addition to fluctuating timber prices, natural hazards such as storm, snow break or insect attacks were used following the model by Clasen et al. (2011). These risks burden every forest enterprise by creating extraordinary expenses more or less randomly and thus leading to lower profits, and sometimes even to deficits. To modify the net revenues to include hazard risks, failure probabilities published by Beinhofer (2007, 2010) were used in this model. These power functions were compiled from different empirical surveys from Germany and can be used to estimate the probability of a particular stand of spruce or beech being completely destroyed at a specific age (Figure 2).

Figure 2 shows that there were similar failure probabilities for the two tree species in younger age-classes. However, when 100-year-old forest stands were compared, beech had a much lower cumulative failure probability of 10 per cent in contrast to

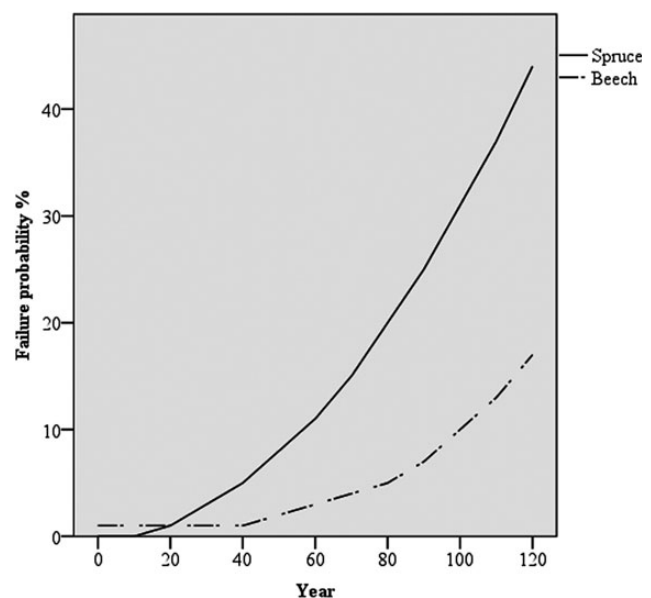


Figure 2 Stand failures of spruce and beech.



spruce with 31 per cent. Where damage to the timber was simulated, harvest revenues were reduced by half (Dieter, 2001). Bright and Price (2000) justified this reduction due to falling prices for wood, reduced quality or additional costs for harvesting. Furthermore, in the case of a hazard, replanting of the stand was simulated, but with lower costs in older stands, as existing regeneration was assumed.

For both tree species, the same risk conditions for all growth regions were assumed, so that possible deviations in required compensation payments could mainly be explained by different site conditions. In so doing, a Monte Carlo simulation (Barreto and Howland, 2006) with 10 000 repetitions for each growth region and tree species was carried out to generate annuities and their standard deviations. Following the method used by Clasen *et al.* (2011), it was then calculated a compensation amount – in case of homogenization – which would be required to make the increased risk bearable. This approach is based on the theory of Markowitz (1952), who showed that a portfolio of mixed assets induces lower risks compared with the summation of each single risk. Based on this, the so-called capital market line can be calculated (Tobin, 1958), which can show the risk–return relationship for every efficient portfolio mixture. This relationship is also known as the Sharpe-Ratio (Sharpe, 1966). Following these considerations, Clasen *et al.* (2011) showed, if a certain mixture of tree species was chosen based on the acceptable level of risk, but subsequently a pure forest developed unintentionally (e.g. due to game browsing), thus creating a higher risk, then the original risk–return relationship of a mixture could be reached with risk compensation. These calculations are possible for every species composition and diminishing recruitment scenario. For our considerations of all growth regions, initial mixtures of both 30 and 50 per cent admixed beeches were considered, with subsequent homogenization to pure spruce forest.

## Results

The results are divided into two parts. First, the validation of site-specific growth will be shown, since the financial results are based on this variation for both tree species. Second, the impact of site conditions on financial result will be demonstrated. On the one hand, the impact on annuities will be demonstrated. On the other hand, it will be shown which influence different site conditions will have on the level of risk compensation required due to homogenization of tree species.

### Standing volume under different site conditions

As expected, spruce achieved the highest standing volume at the end of the rotation period, which was, on average,  $604 \text{ m}^3 \text{ ha}^{-1}$  (gross volume under bark) (for numeric representation, see Tables A1 and A2). However, the range of standing volumes over all growth regions showed great variability, with a standard deviation of  $\sim \pm 48 \text{ m}^3 \text{ ha}^{-1}$ . Spruce stands in the foothills of the Alps achieved the highest value ( $656 \text{ m}^3 \text{ ha}^{-1}$ ), in contrast to the growth region *Untermainebene* – an outlier on the very low end of the volume range ( $408 \text{ m}^3 \text{ ha}^{-1}$ ). Therefore, in cooler mountain regions, spruce growth achieved significantly higher standing volumes. In contrast, variation in beech volume at the end of rotation period was significantly lower –  $\pm 19 \text{ m}^3 \text{ SD}$  per hectare,

with an average volume of  $417 \text{ m}^3 \text{ ha}^{-1}$ . Here, the lowest value occurred in northern Bavaria (*Fränkischer Keuper and Alpvorland*),  $387 \text{ m}^3$ , while the highest value occurred in southern Bavaria,  $446 \text{ m}^3$ .

### Individual growth in comparison with yield tables

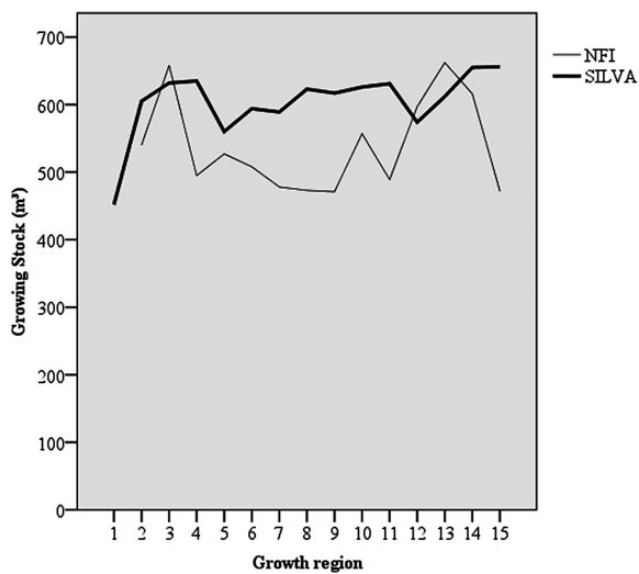
Yield tables include the most important numerical values for evaluating forest stands. They are considered valid for larger spatial units, and regional variances are generally not accounted for. Further, yield tables reflect productivity of stands at the time of their construction. Therefore, a change in productivity due to anthropogenic or climatic effects is also usually not considered, although it has already been observed in many places (Pretzsch, 1999). Nevertheless, yield tables can be used to determine the range of possible site quality values for the simulated data in order to validate the average volumes for each growth region produced by the simulation. For example, a calculated volume representative of a very low site quality for spruce in the low mountain ranges would not be realistic.

Some German yield tables do include site productivities for Bavaria. Vanselow (1951) documented spruce growth in southern Bavaria. Our simulated average volume in southern growth regions of between  $612$  and  $656 \text{ m}^3$  (gross volume under bark) at the age of 100 corresponds with a site class between 2 and 2.5. For the middle and northern regions of Bavaria, yield tables by Wiedemann (1936) and Gehrhardt (1923) are quite suitable. The different timber volumes calculated in our analysis for these regions at the age of 100 could thus be classified between site class 1 and 2 (Wiedemann, 1936) and between 1.5 and 2 (Gehrhardt, 1923). The growth region *Untermainebene* produced yields corresponding to comparatively lower site average quality indexes of  $\sim 2.5$  (Wiedemann, 1936) or 3 (Gehrhardt, 1923). For beech, the average timber volume of between  $386$  and  $446 \text{ m}^3$  simulated by our study across all growth regions would be classified as representative of site class ranges from  $\sim 1.5$  to 2.5 (Wiedemann, 1936) or from 2.5 to 3 (Gehrhardt, 1923) at age 120.

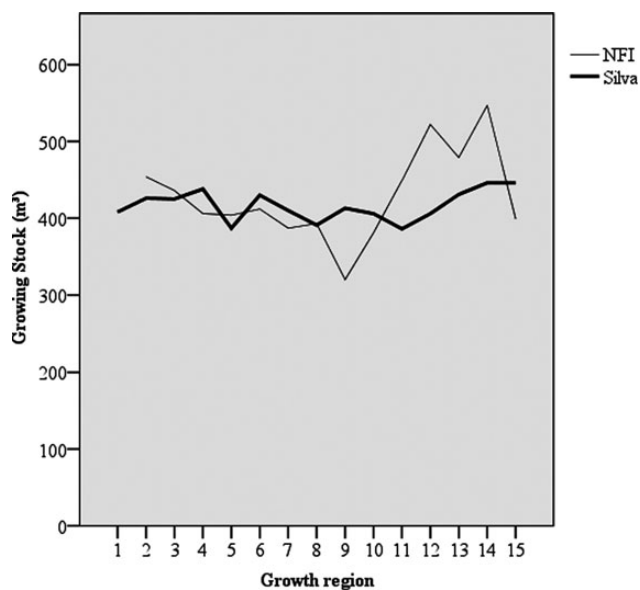
### Individual growth in comparison with NFI 2

The NFI is a large-scale survey of the German forest. The information it provides is derived from a 4 by 4 kilometre sampling plot grid, whereby regional evaluations are possible (Polley and Kroiher, 2006). For validation of the data simulated using SILVA 2, timber volume at the age of 100 for spruce and 120 for beech for each growth region was compared (Figures 3 and 4). However, the NFI volume figures are not ideal for validating the volumes derived from SILVA 2. The smaller the growth region – and therefore the sample size in the NFI survey – the less reliable the information (the sample plot grid in growth region one, for example, is not sufficient for this comparison). Thereby, a proportion of 20 per cent of admixed tree species to increase the sample size of ‘pure’ stands was tolerated. Furthermore, the NFI collects data for age-classes only (Polley *et al.*, 2010). For spruce, for example, data for the two age-classes, 80–100 and 100–120, are recorded for the NFI. For our study, these were averaged to get specific timber volume in a growth region for individual 100-year-old ‘pure’ spruce stands.

Figure 3 shows standing volumes in cubic metres per hectare (gross volume under bark) for spruce. In most cases, NFI volume was lower than that derived through SILVA simulations. This is due



**Figure 3** Comparison of spruce timber volume ( $\text{m}^3$  under bark) at the age of 100 between NFI Data and SILVA simulation in 15 growth regions.



**Figure 4** Comparison of beech timber volume ( $\text{m}^3$  under bark) at the age of 100 between NFI Data and SILVA simulation in 15 growth regions.

to the fact that SILVA 2 does not consider any storm or bark beetle attacks that decrease average timber volume (Hahn and Knoke, 2010). These effects were instead considered in the financial valuation portion of the analysis. However, except in southern Bavaria, beech growth modelled by SILVA 2 was more or less comparable with NFI data (Figure 4), since beech is less vulnerable to calamities. For the highly favourable growing conditions of southern Bavaria, however, the SILVA 2 growth simulation was not able to generate beech growth comparable with that shown in the NFI data.

### Individual growth in comparison with site quality risk modelling

Kölling *et al.* (2009, 2010) modelled site quality-dependent risk profiles for tree species in Bavaria. Hereby, average annual temperature and precipitation, as well as hydrologic balance for each growth region, were considered (year 2000 as reference). The authors characterized seven risk profiles. Considering current site conditions, the results for spruce and beech were as follows: Kölling *et al.* (2009, 2010) classified spruce growth in growth region one as having high risk, in region four as significantly increased risk, and region five as medium–high risk. They classified four regions as having recognizable risk (three, seven, nine and twelve) and two regions as low risk (six and ten). The authors classified all other regions as having very low risk for spruce growth. When it was performed a correlation between the site quality differences simulated by Kölling *et al.* (2009, 2010) and the growth simulation results obtained by SILVA 2, some correspondence was recognizable: growth region one (Untermainebene) showed by far the most unfavourable growth conditions for spruce. Spruce growth in the northern and low altitude parts of Bavaria was also poor. However, good growth was present in both the low mountains and the alpine region. Beech, on the other hand, showed a very low risk for all growth regions under the simulation by Kölling *et al.* (2009, 2010). This tendency for beech was also shown by the SILVA simulation with more or less equal timber volumes across all regions (see Table A2).

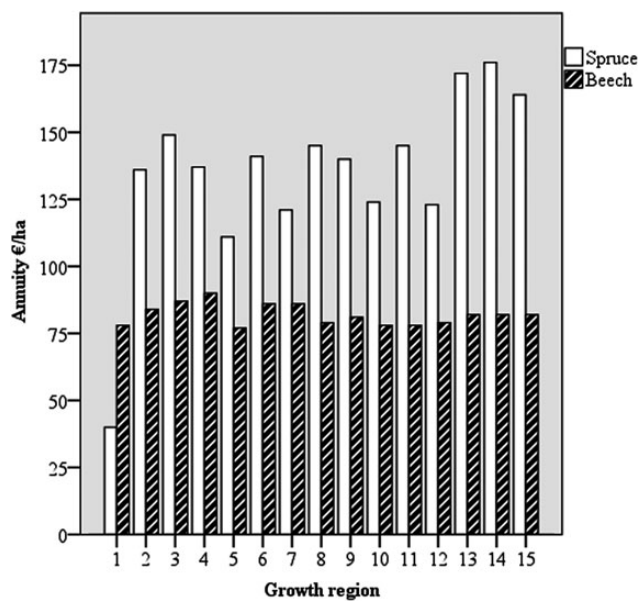
### Financial growth site variations

With regard to the stumpage values that were obtained at the age of 100 for spruce and 120 for beech, the following could be seen: the average value for spruce for all growth regions was  $\sim 28\,671 \text{ € ha}^{-1}$ , with a minimum of  $12\,356 \text{ € ha}^{-1}$  and a maximum of  $32\,795 \text{ € ha}^{-1}$ . Beech values showed less variation, with a minimum of  $14\,793 \text{ € ha}^{-1}$ , a maximum of  $19\,132 \text{ € ha}^{-1}$ , and an average of  $\sim 16\,653 \text{ € ha}^{-1}$ .

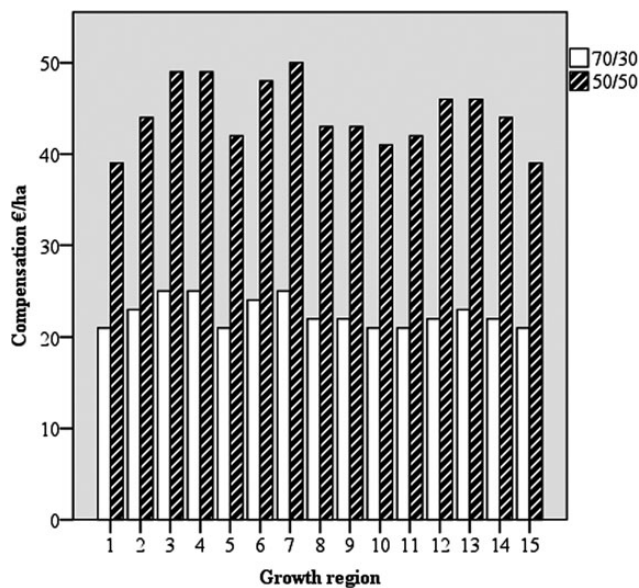
Considering all financial flows during the whole stand life of spruce and beech, the resulting calculated annuities were as in Figure 5: again, spruce showed significant deviations. With all payments caused by thinnings included, the arithmetic mean annuity value was  $\sim 135 \text{ € ha}^{-1}$ . However, the values range from 40 to  $176 \text{ € ha}^{-1}$ , with a standard deviation of  $\pm 32 \text{ € ha}^{-1}$  (see Table A2). Even when the outlier (Untermainebene) was excluded, the standard deviation was still  $\sim 19 \text{ € ha}^{-1}$ . Favourable site conditions for spruce – cool and humid mountain regions – were reflected in higher annuities, as seen for low mountain regions 3 and 11 and alpine regions 13, 14 and 15. By comparison, the more or less equal growth for beech in all growth regions was reflected in the small range of the annuities as well ( $77\text{--}90 \text{ € ha}^{-1}$ ), with a relatively small standard deviation ( $\pm 4 \text{ € ha}^{-1}$ ).

### Risk payment variations

Figure 6 illustrates two scenarios. The two starting points were a mixture of spruce and beech from natural regeneration in a ratio of 70/30 and a ratio of 50/50. A complete recruitment failure of beech due to ungulate browsing over an extended period of time was assumed. The consequence was thus an evolution into a pure stand of spruce for both scenarios. Recruitment failure of



**Figure 5** Annuities of spruce and beech stand life depending on growth site.



**Figure 6** Compensation payments in € ha<sup>-1</sup> for two mixtures of spruce and beech starting with 70 and 50% spruce with following loss of beech for 15 growth regions of Bavaria each.

the initial 30 per cent component led to lower risk compensation than recruitment failure of the 50 per cent beech component. The higher the recruitment failure of an initial mixed stand, the more the effect of site variations is perceivable. The respective payments for each of the growth regions for the initial 30/70 mixture fluctuated between 21 and 25 € ha<sup>-1</sup>. If admixture of 50 per cent beech is impossible, payments between 39 and 50 € per year and hectare are required. For both scenarios, the relationship between the differences in payments among the various regions was approximately the same. Growth regions three, four, six and seven required the highest payments for both scenarios. On first

examination, clear relationships between annuity deviations (Figure 5) and compensation payments (Figure 6) could not be derived. However, the predictor was found to be the respective individual return–risk ratio (Sharpe-Ratio (Sharpe, 1966)) for each growth region, which was reflected in the site variations of each single period. Here, one of the key issues in this context was the beech annuity – the higher the beech annuity, the higher the compensation. Thus, growth regions three, four, six and seven tended to the best annuity – if only slightly. The size of the spruce annuity, however, had only a minor influence on this calculation (see growth region one). Another cause could be the risk of spruce. Although the growth in the regions was simulated with the same conditions for both timber fluctuations and survival probabilities, the risks differed in relationship to the returns, as the timber volumes of the single periods were different.

## Discussion and conclusion

This study illustrates the effectiveness of using average site conditions over large regions to estimate financial output, as well as the degree to which site variations affect required compensation payments due to ungulate browsing. The following phenomenon was observed: spruce growth showed higher variation between growth regions than beech. Compensation payments fluctuated among growth regions, and these fluctuations were mainly correlated with the amount of the beech annuity and the level of risk for spruce. These findings are quite understandable since in this calculation it was asked for recruitment failure of an admixed tree species to be compensated and about the risk of pure stand a forest owner will be taking. Thus, the higher the value of a lost species and the higher the unacceptable risk, the higher the necessary compensation. This analysis also demonstrates that such risk calculations cannot generalize, as varying site conditions also have a considerable influence on financial result. The greater the variation in site conditions, the more they should be considered – particularly in the case of politicized issues such as game browsing.

Considering different growth conditions is an essential part of this study. However, it should mention that this study describes a simulation with average growth parameters. If a growth region consisted of more than one district, one growth district as representative for the whole region as a starting point was used. Also, the choice of average start parameter of the SILVA simulation (mean diameter and basal area) allows certain flexibility. Nevertheless, the effects of growth site variations are clear, and the trends shown indicate the influence they can have on economic calculations. Rough estimation of stand structures at the regional level (Dursky, 2002) and a growth simulation up to rotation period (Pretzsch, 2002) both are well-known methods.

The consistency of beech growth across different site conditions and contrasting variation in spruce were also reported by Rötzer *et al.* (2010). Naturally, beech is very tolerant to different site conditions (Pretzsch and Dursky, 2002), whereas lowlands with high temperatures and relatively little precipitation lead inevitably to lower growing stock for spruce (Pretzsch, 2003). This effect is no surprise given the ecogram of the tree species. Here, beech shows its high ecological amplitude to dominate natural vegetation, whereas spruce is generally considered to be a tree species of the mountains (Ellenberg and Leuschner, 2010).

The financial consequences of different regional growth potentials can be shown well with stumpage values and annuities. For



each rotation period, the median values, which were calculated for the 15 growth regions (Table A2) – for spruce (~30 000 Euro) and beech (~17 000 Euro) – are in line with other economic analyses, although these studies were based on different assumptions of timber quality or timber prices. The stumpage value for spruce was calculated at ~30 000 € at the age of 100 by Hanewinkel *et al.* (2010), ~33 000 € at the age of 140 by Dieter (2001) and ~24 000 € at the age of 112 by Clasen *et al.* (2013). Regarding beech stumpage with length of rotation periods similar to those used in our simulation, values for beech calculated by other studies were lower than those for spruce as well – ~25 000 € calculated by Hanewinkel *et al.* (2010) at the age of 120 and ~22 000 € calculated by Dieter (2001) at the age of 140. Overall, the difference in the expected financial output between spruce and beech has been estimated at 60 per cent (Möhring, 2004). Considering beech annuities, the calculated values were, on average, 61 per cent of average spruce values (82 € ha<sup>-1</sup> in relation to 135 € ha<sup>-1</sup>, see Table A2).

Garcia-Gonzalo *et al.* (2007) indicated that empirical models, such as growth simulations, which are based on experimental plots, exclude the impacts of climate change on forest growth. Thus, they assume that future environmental conditions will be similar to those in the past. It is, however, crucial to estimate tree growth in the future due to the long rotation periods of spruce and beech forests. In this context, Kölling *et al.* (2009, 2010) predicted possible changes in temperature and hydrologic balance for Bavaria up to the year 2100 (regional climate scenario B1, WETTREG). The changes are remarkable. Only the alpine region showed a low risk for spruce growth. Six growth regions showed a high risk for spruce and a very high risk arose in growth regions one (*Untermainebene*) and four (*Fränkische Platte*). Therefore, an acceptable level of risk for spruce would only be possible in the mountain regions. Looking at our data as an artificial time series for economic consequences from expected environmental changes, the following can be expected due to these changes: average spruce income for six growth regions will be the same as that found for the *Untermainebene* region given current conditions. This means that spruce can be expected to generate less income than beech in these regions in the future. Income expectations for spruce in two growth ranges are even lower. In comparison, the risk for beech will remain in the same risk class in 11 of the 15 growth regions. Only the *Untermainebene* will increase to very high risk for beech. On average, spruce growth will turn from low risk to significantly increased risk for the whole of Bavaria, whereby beech will keep the same risk profile and therewith more or less constant income possibilities in relationship to site conditions.

Regarding possible climate changes, the following can be deduced for calculating risk payments: in consideration of the results of Kölling *et al.* (2009, 2010), the risk for spruce will increase with associated higher site variations. In this study, a constant survival probability for all growth regions was assumed to investigate the influence of site variations. However, varying forest site conditions also influence the survival probability of tree species – in particular in the case of increasing drought. With a higher risk for spruce in the future, the risk of a loss of beech in a mixed stand with spruce would require even higher compensation levels. With a higher variability between the growth regions, a higher variation in risk compensation levels can thus be expected.

It can be concluded, in addition to the well-known risk factors for timber price fluctuations or occurrences of calamities, variable

growth potential in different regions also implies a substantial degree of financial variation. This consideration is particularly critical in order to draw economic conclusions from studies that are based on specific site conditions for a single growth region, especially in calculating ungulate damage and especially in consideration of climate change.

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## Conflict of interest statement

None declared.

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Table A2 Standing stock (m<sup>3</sup> of harvest per hectare) at rotation period

| Tree species                                                    | Arithmetic mean | Median | Minimum | Maximum | Standard deviation |
|-----------------------------------------------------------------|-----------------|--------|---------|---------|--------------------|
| Spruce                                                          | 604             | 617    | 452     | 656     | 50                 |
| Beech                                                           | 417             | 413    | 386     | 446     | 20                 |
| Financial value of final standing volumes (€ ha <sup>-1</sup> ) |                 |        |         |         |                    |
| Spruce                                                          | 28 671          | 30 214 | 12 356  | 32 795  | 4892               |
| Beech                                                           | 16 653          | 16 554 | 14 793  | 19 132  | 1218               |
| Annuity for different site conditions (€ ha <sup>-1</sup> )     |                 |        |         |         |                    |
| Spruce                                                          | 135             | 140    | 40      | 176     | 32                 |
| Beech                                                           | 82              | 82     | 77      | 90      | 4                  |

Appendix

Table A1 Growing stock (m<sup>3</sup> of harvest ha<sup>-1</sup>) at rotation period

| Growth region                                     | Spruce | Beech |
|---------------------------------------------------|--------|-------|
| 1. Untermainebene                                 | 452    | 408   |
| 2. Spessart-Odenwald                              | 605    | 426   |
| 3. Rhön                                           | 632    | 425   |
| 4. Fränkische Platte                              | 635    | 438   |
| 5. Fränkischer Keuper                             | 560    | 387   |
| 6. Frankenalp/Oberpf. Jura                        | 594    | 430   |
| 7. Fränkisches Triashügelland                     | 589    | 410   |
| 8. Frankenwald, Fichtelgebirge, Steinwald         | 623    | 391   |
| 9. Oberpfälzer Becken- and Hügelland              | 617    | 413   |
| 10. Oberpfälzer Wald                              | 626    | 406   |
| 11. Bayerischer Wald                              | 631    | 386   |
| 12. Tertiäres Hügelland                           | 574    | 406   |
| 13. Schwäb. Bay. Schotterpl./Altmoränenlandschaft | 612    | 431   |
| 14. Schwäb.-Bay. Jungmoräne and Molassevorberge   | 655    | 446   |
| 15. Bayerische Alpen                              | 656    | 446   |